Vector Short Listing Test – JANUARY 2019

Module : C / MAX Marks : 25

Name :	**Marks Obtained:**
Vector ID :	

1. What is the output of the below code.
#include<stdio.h>
main( )
{ int a = -3;
a = -a -a + !a;
printf("%d",a);
}
	Ans:

2. main()
{ float a = 123.4, b = 456.8;
a ^= b ^= a ^= b;
printf(" %d t %d" , a, b);
}
	Ans:

3.main()
{
int k, num=30;
k=(num > 5 ? (num <=10 ? 100:2000):500);
printf("%d",k);
}
	Ans:

4. main()
{
int i,j; i=j=2,3;
while(--i&&j++)
printf("%d %d",i,j);
return 0; }
	Ans:

5. #include<stdio.h>
int main()
{ void *vp;
char ch=74, *cp="JACK";
int j=65;
vp=&ch;
printf("%c", *(char*)vp);
vp=&j;
printf("%c", *(int*)vp);
vp=cp;
printf("%s", (char*)vp+2);
return 0;
}
	Ans:

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6. Scope of a global variable which is declared as _____ is within the file.

7.
```
void print(char *);
main(){
        char str[]="Geneius";
        print (str);
}
void print(char *s){
        if(*s) print(++s);
        printf("%c",*s);
}
```

8. The parameters in a function definition are_____

9. Even if integer / float arguments are supplied at command prompt they are treated as _____.

10.
```
main(){
        int arr[2][3][2]={{{2,4},{7,8},{3,4},}, {{2,2},{2,3},{3,4}, }};
        printf(" %d",**(*arr+2)+2+7);
}
```

11.
```
main( ){
        static int a[ ] = {0,1,2,3,4};
        int *p[ ] = {a,a+1,a+2,a+3,a+4};
        int **ptr = p;
        ptr++;
        printf(" %d %d %d", ptr-p, *ptr-a, **ptr);
        *++ptr;
        printf(" %d %d %d", ptr-p, *ptr-a, **ptr);
}
```

12.
```
main()
{ union
{ struct
{ char c[2];
 char ch[2];
}s;
struct
```

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```
{ short int i;
short int j;
}st;
}u={ 12,1, 15,1};
printf("%d %d",u.st.i, u.st.j);
}
```

13. What is the value of CAR in the following statement?

```
enum vehicle{ BUS, SCOOTER=2,CAR,TRAIN=5,AEROPLANE=6}
```

14. What is the size of ptr1 and ptr2 ?

```
struct x {
        int j;
        char k[100];
        unsigned I;
        };
        int *ptr1; struct x *ptr2;
```

15. The nodes in a linked list are implemented using_____

16.

```
#define big(a,b) a > b ? a : b
#define swap(a,b) temp=a; a=b; b=temp;
void main() {
        int a=3,b=5,temp=0;
        if ((3+big(a,b)) > b) swap(a,b);
        printf("%d %d",a,b);
}
```

17. If the binary tree is having only root node, then the height is _____

18. If front=rear = -1 that means queue is _____

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Vector Short Listing Test – JANUARY 2(

Module : C / MAX Marks : 25

19.A double data type always occupies _____ bytes.	Ans:
20. A function assumes _____ data type as return type if no return type is specified in the funtion prototype.	Ans:
21. Write the c program to implement the user defined function for atof.(5 m)	Answer: W the White

Answer Question No. 21 Here

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Vector Short Listing Test – JANUARY 2019

Module : CPP / MAX Marks : 25

Name :	Marks Obtained:
Vector ID :	

1. _____ operator is used to extract the data from file?	Ans:
2. Which stream class is to only to write to the files ?	Ans:
3. Write the syntax for preincrement operator overloading using friend function	Ans:
4. A virtual function that has no definition within a base class is called___	Ans:

5.What is the output of the below code?
include<iostream>
using namespace std;
class Base{
public: virtual void show() = 0;
};
class Derived : public Base
{
 	
};
int main(void) {
Derived q;
return 0;
}
	Ans:

6. A protected member inherited in public mode becomes _____ in the derived class	Ans:
7. main() { int a=9; int k=10; int &r=a; cout<<r ; r=k; cout<<r ; } What is the output of the above code?	Ans:
8. The mechanism of deriving a class from another derived class is know as _____	Ans:

9. int sum(int a, int b=5, int c=10);
main() {
cout<<sum(5)<<endl<<sum(10,5)<<endl<<sum(5,10,10);
}
int sum(int a, int b, int c) {
	Ans:

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Vector Short Listing Test – JAN

Module : CPP / MAX Mark

```
    return a+b+c;
}
```

What is the output of the above code?

10. class that contains at least one pure virtual function, is said to be_____

11. _____ means that both the data and the methods which may access it are defined together in the same unit.

12. A function call resolved at the runtime is referred to as _____ binding

13. The _____ of a function is not a factor in distinguishing overloaded functions.

14. The feature that allows you to use the same function name for separate functions that have different argument lists is called _____

15. The exception is caught & processed using____

16. During a class inheritance in CPP, if the visibility mode or mode of derivation is not provided, then by default visibility mode is _____.

17. The private data of a class can be accessed only through the _____ and _____ of a class.

18. Which keyword will be used to refer to the current object ?

19. Every class has at least one constructor function, even when none is declared.

20. The operator allows to define the member functions of a class, from outside the class?

21. What is the size of wchar_t in C++

22. If I have same variable name for both local and global variable , then how to access the global variable?

23. Write the syntax of new for allocating memory for 1-D array

24. A generic function that defines a general set of operations that will be applied to various types of data, is created using the keyword

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25. What is the output of below code?

```
int &fun() {
        static int x = 10;
        return x;
}
int main(){
        fun() = 30;
        cout << fun();
        return 0;
}
```

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Vector ID :

1. Define Virtual memory. Why it is useful in Linux? (2)
2. Why virtual memory is not preferred in RTOS? (2)
3. What is deadlock? List main causes of deadlock. (2)
4. What is IPC. List the IPCs in linux. (2)
5. What is a zombie job? How to avoid zombie? (2)
6. What are the advantages of sigaction over signal. (2)
7. List the use of fcntl(). (2)
8. differentiate between semaphore and mutex. (2)
9. Define response time, turnaround time, throughput (3).
10. what is inode ? List the members(contents) of inode(3).
11. List the types of files in Linux. (3)

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Vector Short Listing Test – January 201

Module : TCPIP / MAX Marks : 25

	Marks Obtaine
Name :	
Vector ID :	

1. Identify the class of following IP address: 4.5.6.7.	Ans:
2. The loop-back address is used to send a packet from the ____ to ___ on the network.	Ans:
3. What destination address can be used to send a packet from a host with IP address 188.1.1.1 to all hosts on the network?	Ans:
4. Repeaters function in __ layer of OSI?	Ans:
5. The host byte order for sending and receiving information using TCP/IP protocol suite can be _____ endian.	Ans:
6. As the data packet moves from lower to upper layers, headers are___.	Ans:
7. What is the main function of the transport layer?	Ans:

Note : Answer the following questions on a separate sheet

8. In a client-server model for connection-oriented communication, the order of the use of socke server should be :

ANS: (2 Marks)

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Vector Short Listing Test – January 2019

Module : TCPIP / MAX Marks : 25

9. Fragmentation and Reassembly is done by _____ protocol in _____ of TCP/IP layered architecture.

(2 Marks)

10. IPV4 address is _____ bytes and represented as _____ number for example 1.2.3.4

(2 Marks)

11. The SFD field in ETHERNET FRAME is _____ byte, in which last two bits are _____. (2 Marks)

12. The Router _____ the TTL value in an IP datagram before forwarding it to the next router. But if TTL value become _____ after decrementing, then that IP datagram is discarded. (2 Marks)

13. The range for the private IP addresses for class A is _____ to _____. Hence total number of network blocks are _____. (3 Marks)\

14. Lists 5 connecting devices in increasing order functionality. (5 Marks)

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1. UART protocol was conceived by_____.(1M) Ans:
a. Motorola b. Philips c. EIA d. none

2. Which nibble of the TCON is used for Timer functionality? (1M) Ans:

3. Interrupt vector address for UART interrupt is _____. (1M) Ans:

4. In mode 0, the TF0 flag sets when the timer rolls over from ___to ___.(1M) Ans:
a. 0xffff to 0x0000 b. 0xff to 0x00 c.0xff1f to 0x00 d. none

5. For an 8051 operating with a 6 MHz crystal frequency, a 4 machine cycled instruction takes ___us to execute.(1M) Ans:

6. In I2C protocol the serial data transfer is _____ first. (1M) Ans:

7. The size of "mov 30h,20h" instruction is_____ byte(s).(1M) Ans:

8. Upon power up, the program counter (PC) has a value of _____.(1M) Ans:
a. 0x0000 b. 0x0007 c. 0x0ff d. 0xffff

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Vector Short Listing Test – January 2019

Module : 8051 / MAX Marks : 25

9. How many nibbles will make 16 bits? (1M) Ans:

10. SPI Protocol supports how many operating modes ?(1M) Ans:

11. Maximum, how many numbers of slaves can be connected on I2C bus? (1M) Ans:

12. mnemonic used to exchange the nibbles within the same operand is _____.(1M) Ans:

a. SWAP b. XCH c. XCHD d. MOV

13. Keil C51 supports _____ memory models.(1M) Ans:

14. Which timer of the 8051 is used to set the baud rate for UART communication?(1M) Ans:

15. Explain any 2 addressing modes of 8051 with examples?(2M) Ans:

16. Draw the timing diagram for I2C bus events i.e. START and STOP? (2M) Ans:

17. Write I2C byte read algorithm?(3M) Ans:

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Vector Short Listing Test – January 201

Module : 8051 / MAX Marks : 25

Ans:

18. Baud rate formulae in UART standard mode _____.(1M)

19. Draw the embedded C program for generating 1msec time delay using timer0?(3M) Ans:

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